Abstract*Abstract.

Do Standard Neural Networks Learn par Block-Spin Renormalization Group Transformations?

A Controlled Negative Result on the 2D Ising Model

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Abstract: We investigate whether standard neural networks (MLP with GELU activations) trained on physics problems exhibit *RG-like behavior*, using the 2D Ising block-spin transformation as a controlled test. The main positive finding is that MLPs trained with block-spin labels achieve test MSE 1.26 ± 0.65, notably lower than the random-label baseline (1.81 ± 0.02, $n = 10$ seeds; 95% bootstrap CI for the mean difference $[0.31, 0.79]$ excludes zero, satisfying Criterion R1). The main negative finding is that a single linear layer achieves 0.20 ± 0.11, substantially outperforming the MLP (1.26 ± 0.65) on the same task, with all 10 linear seeds lower than all 10 MLP seeds. This demonstrates that the block-spin target at $L = 16$ is approximately linear in the spin variables; the MLP's non-linear hidden layers provide no advantage — and considerable instability (seed variance 0.65) — on this task. Cross-scale transfer (train at L, test at L') was measured for the first time: both MLP and linear models degrade with scale distance, but neither outperforms the linear baseline on this criterion, leaving R2 undecided. A turbulence extension (2D LBM, Re= 200) yields $\alpha = -1.30 \pm 0.05$ vs. Kolmogorov $-5/3 \approx -1.67$, and is explicitly disclaimed as not constituting evidence for any criterion. This paper's central contribution is a *scoped* negative result: standard deep architectures do not exhibit RG-like advantages on block-spin tasks that are well-approximated by locally linear projections. This does not contradict prior RBM-based findings (Mehta & Schwab, 2014), which rely on an explicitly RG-structured inductive bias absent from MLPs.

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0.1 Introduction

The renormalization group (RG), developed by Wilson [23] and crystallized in Kadanoff's block-spin picture [10], describes how physical systems behave across scales. At a critical point, the RG transformation converges to a fixed point; the approach to this fixed point is governed by critical exponents [7, 25].

Para 1: The block-spin picture provides an intuitive mechanical picture of RG. Imagine a lattice of spins, each pointing either up (+1) or down (-1). The RG transformation groups nearby spins into blocks — for a 2×2 block size Λ, four adjacent spins become one coarse-grained spin. To determine the new spin, we take a majority vote: if three or four spins in the block point up, the coarse-grained spin points up; if three or four point down, the coarse-grained spin points down. This reduces an $L \times L$ lattice to an $L/2 \times L/2$ lattice. Repeating this operation at multiple scales generates the RG flow. At each step the system is described by fewer degrees of freedom but should represent the same physics at longer wavelengths.

Para 2: At the *critical temperature* β_c, something remarkable happens: fluctuations occur at *all* scales simultaneously. A spin at the critical point is correlated with other spins at all distances — correlation functions decay as a power law, not exponentially. There is no characteristic length scale in the system (the correlation length ξ diverges to infinity). This is why the critical point is where RG fixed points live: at β_c, the system looks statistically the same no matter what magnification you use. Zoom in on a critical system and you see the same pattern of large and small fluctuations you saw before zooming.

Para 3: The "fixed point" in RG language is a system that is *invariant* under the coarse-graining transformation. If you apply the block-spin transformation to a system already at the fixed point, you get back a system with the same statistical properties — same correlation functions up to a rescaling of length. The fixed point is the intersection of all RG flows starting from different microscopic parameters (different temperatures, fields, couplings). Physical quantities near the fixed point obey power laws; the exponents of these power laws are the critical exponents. They are universal: they depend only on the dimensionality and symmetry of the order parameter, not on the microscopic lattice details.

Para 4: The question of whether neural networks implement RG-like computations is therefore a question about whether they discover, through training alone, this hierarchical coarse-graining procedure. If a standard architecture trained on physical data implicitly learns to block-average across scales, this would constitute a strong inductive bias toward the kind of scale-invariant reasoning that RG formalizes.

The hypothesis that neural networks might *implement* RG-like computations was proposed by [16], who observed that restricted Boltzmann machines learn effective couplings matching RG flow in the 2D Ising model. Subsequent work formalized this as "Neural RG" [18, 14, 12], with explicit RG-motivated architectures.

We take a different stance: can *standard* architectures (MLP with GELU, no explicit RG structure) learn block-spin transformations, and do they show quantifiable signatures of RG-like behavior? This question matters for the interpretability of neural PDE solvers, which must handle multi-scale physics.

The broader significance extends beyond the Ising model. If standard neural networks already implement RG-like computations, this would partially explain their striking success in modeling multi-scale physical systems such as turbulent flows [19, 5], crack

propagation in heterogeneous materials [6, 20], and material defect dynamics. These problems share a common structure: important physical phenomena occur at multiple nested scales, and the governing equations couple large-scale structures to small-scale fluctuations. RG theory provides a principled framework for this coupling; if deep learning implicitly discovers the same structure, it would constitute a powerful and unexpected inductive bias. Conversely, if standard architectures do *not* exhibit RG-like behavior, this raises the fundamental question of what *other* inductive biases they do rely upon — and whether those biases are sufficient for robust extrapolation in physical science. This connects to the broader research program of "machine learning meets theoretical physics" [16, 18, 14, 12], which seeks to understand the relationship between the structure of physical law and the structure of learned representations.

Research Questions

1. Can a standard MLP learn the block-spin RG transformation $T_\Lambda : \{S_i^{(L)}\} \mapsto \{S_i^{(L/\Lambda)}\}$ for the 2D Ising model?

   Here, "learn" means *supervised regression*: we generate examples of fine-grained spin configurations S^{fine} along with their coarse-grained labels S^{coarse} computed via the majority-vote block-spin transformation (Eq 3), and train the network to predict S^{coarse} from S^{fine}. This is not an unsupervised flow-based model nor a generative model; it is a direct function approximation task. The question is whether a standard MLP architecture can represent the block-spin map without having it hardwired into the architecture.

2. Is this learning statistically meaningful (vs. random baselines)?

   Any sufficiently expressive model can memorize a finite training set. To distinguish genuine block-spin learning from mere memorization, we compare against a *random-label baseline*: the same MLP architecture trained on the same fine-grained inputs but with labels randomly shuffled (each coarse-grained spin independently assigned ± 1 with probability 1/2). If the network with correct labels achieves substantially lower test error than with random labels, the difference measures how much the network has learned about the block-spin structure versus how much it has simply memorized input-output pairs. This baseline is standard in deep learning theory [24] and is essential here because the Ising spin configurations at β_c contain rich structure that a high-capacity network could in principle fit to arbitrary labels.

3. Does the learning quality degrade with scale distance, as RG theory predicts?

   RG theory's central quantitative prediction is that information about the microscopic scale is progressively lost under repeated coarse-graining. If a network is trained on block-spin targets at one lattice size L and then tested at a different size L', the test error should increase monotonically with $|L - L'|$, because the RG flow has moved further away from the training scale. This cross-scale transfer is *the* defining operational test of RG-like behavior in our framework. It is qualitatively distinct from within-scale learning (which merely confirms the network can represent the block-spin map at a single scale) and requires the network to have discovered the *hierarchical* structure of scale-coupling, not just a local majority-vote rule.

4. Does the framework extend to a neural turbulence closure reproducing Kolmogorov spectrum scaling?

Kolmogorov's 1941 hypothesis [13] for fully developed turbulence states that the energy spectrum in the inertial range follows $E(k) \propto k^{-5/3}$, where k is the wavenumber. This power law emerges from a cascade: large eddies inject energy at the forcing scale, break into smaller eddies, and so on until viscosity dissipates at the Kolmogorov microscale. The $-5/3$ exponent is remarkable because it can be derived almost entirely from dimensional analysis plus the assumption of universality (the inertial range statistics depend only on the energy cascade rate ϵ). If a neural network trained *only* on subgrid stress closures (the small-scale fluctuations unresolved by the simulation grid) can reproduce this exponent, it would constitute strong evidence that the network has learned an RG-like scale-coupling in a physically non-trivial setting beyond the controlled Ising testbed. This would be striking because the network never sees the inertial range directly during training; the correct exponent would have to emerge from the physics embedded in the training data.

Operational Definition: "RG-like Behavior"

We say a trained neural network exhibits *RG-like behavior* if it satisfies:

(R1) **Block-spin learning:** Test MSE on block-spin prediction is significantly better than a random-label baseline at the same β.

(R2) **Scale-dependent degradation:** Cross-scale transfer error (trained at L, tested at L') increases monotonically with $|L - L'|$ (RG flow distance), and the network outperforms linear baselines.

(R3) **Turbulence extension:** A neural closure model trained on subgrid stress reproduces Kolmogorov $k^{-5/3}$ inertial-range scaling in the energy spectrum.

All three criteria must be satisfied for a network to be called RG-like. Failure of any criterion is a useful negative result.

What this paper establishes, and what it does not: This paper locates itself as a scoped negative-result paper. R1 is satisfied ($n = 10$ seeds; 95% bootstrap CI for the MLP-vs-random mean difference $[0.31, 0.79]$ excludes zero). R2 (cross-scale transfer) is measured here for the first time; the results are reported but neither the MLP nor the linear model provides a clear advantage, leaving R2 undecided. R3 (turbulence) is a feasibility demonstration only and does not count as evidence for any criterion.

The main negative finding is: a linear model substantially outperforms the MLP on the same block-spin task, implying the majority-vote transformation at $L = 16$ is approximately linear. This is a precise negative result scoped specifically to *locally linear block-spin tasks*. It does not imply that standard deep architectures cannot learn RG-like computations in general; that would require testing on tasks where the RG transformation is genuinely non-linear in the couplings (e.g., non-zero external field, frustrated magnets, or quantum systems). It also does not contradict the RBM-based positive findings of Mehta & Schwab (2014), which rely on an architectural inductive bias — the bipartite energy function $E(v, h) = -b^\top v - c^\top h - v^\top W h$ — that is absent from standard MLPs. The scope of our claim is limited: standard MLPs do not exhibit RG-like advantages on majority-vote block-spin tasks that are well-approximated by linear projections.

0.2 Background: Ising Model and Block-Spin RG

The 2D Ising model on an $L \times L$ square lattice with periodic boundary conditions has Hamiltonian:

$$H = -J \sum_{\langle ij \rangle} S_i S_j - h \sum_i S_i, \qquad S_i \in \{-1, +1\}. \tag{1}$$

We set $J = 1$, $h = 0$ (zero field). The critical inverse temperature is Onsager's exact value [17]:

$$\beta_c = \frac{1}{2} \log(1 + \sqrt{2}) \approx 0.4407. \tag{2}$$

The 2D Ising model holds a special place in the RG literature for three interrelated reasons. First, Lars Onsager's 1944 exact solution is one of the very few non-trivial phase transitions in statistical mechanics that can be solved analytically in closed form. This provides *ground truth* for all relevant quantities: the free energy, correlation functions, and critically, the critical exponents $\nu = 1$, $\eta = 1/4$, $\beta = 1/8$. Any numerical method for studying RG must be tested against these exact values. Second, the 2D Ising model is the *simplest* model with a non-trivial continuous phase transition: it has a paramagnetic phase at high temperature, a ferromagnetic phase at low temperature, and a second-order transition at β_c with well-defined critical exponents. It captures the essential physics of the ferromagnetic-paramagnetic transition without the complications of additional order parameters, anisotropy, or frustration. Third, the 2D Ising universality class is remarkably robust: any spin model with short-range interactions and a discrete symmetry (Z_2 inversion) on a 2D lattice belongs to the same universality class and shares the same critical exponents. This universality means the 2D Ising results generalize broadly, making it an ideal testbed for controlled theoretical experiments [7, 1, 3].

The block-spin RG transformation with block size $\Lambda = 2$ is:

$$S'_{\lfloor i/\Lambda \rfloor, \lfloor j/\Lambda \rfloor} = \mathrm{sgn}\Big(\sum_{k,l \in \mathrm{block}(i,j)} S_{k,l} \Big), \tag{3}$$

i.e., majority vote within each 2×2 block. Iterating Eq (3) generates an RG flow. At β_c the flow converges to the Wilson-Fisher fixed point; away from β_c it flows to ordered or disordered phases.

The majority-vote construction deserves careful explanation. The Ising order parameter is the *magnetisation* $m = \langle S_i \rangle$, a sum over spins that is discontinuous (non-analytic) at β_c in the thermodynamic limit. At the critical point, the block-averaged spin $\sum_{k,l \in \mathrm{block}} S_{k,l}$ is a continuous random variable (it takes values in $\{-4, -2, 0, 2, 4\}$), but the magnetisation of the coarse-grained system is the sign of this sum. The sign function is a non-linear threshold that implements the discontinuous phase transition: below β_c the sum is overwhelmingly positive (ordered phase, all spins aligned), while above β_c the sum fluctuates around zero (disordered phase). At β_c itself, the sum can be positive or negative with roughly equal probability, and the sign function determines whether the coarse-grained block falls in the up or down domain. Geometrically, the sign function partitions the 4-dimensional spin space (± 1 per spin) into two hemispheres separated by the plane $\sum_i S_i = 0$. This is a *linear* threshold in the space of block-summed features, which explains why the majority-vote transformation is well-approximated by a linear model — the non-linearity enters only through the sign function applied to a single linear combination of the four spins in each block.

Critical exponents for the 2D Ising universality class are exactly known: $\nu = 1$, $\eta = 1/4$, $\beta = 1/8$ [17, 11, 21].

Each critical exponent governs a specific physical quantity near β_c. The exponent ν controls how fast the correlation length ξ diverges as the critical point is approached from either side: $\xi \sim |\beta - \beta_c|^{-\nu}$. Physically, near β_c fluctuations become large and long-range; $\nu = 1$ in 2D Ising means this divergence is relatively slow. The exponent η characterises the decay of the spin-spin correlation function at the critical point itself: $G(r) \sim r^{-(d-2+\eta)}$ with $\eta = 1/4$. In 2D this gives $G(r) \sim r^{-1.75}$, a power law slower than the mean-field prediction r^{-1} because strong fluctuations in 2D modify the correlation structure. The exponent β governs the rise of the order parameter (magnetisation) below β_c: $m \sim (-\epsilon)^\beta$ where $\epsilon = (\beta_c - \beta)/\beta_c$ is the reduced temperature. The value $\beta = 1/8$ (much smaller than the mean-field value $\beta = 1/2$) reflects the difficulty of ordering a 2D system against thermal fluctuations. These three exponents are not independent; they satisfy Rushbrooke's relation $2-\alpha = d\nu$, Josephson scaling $\nu d = 2-\eta+\beta(\delta-1)$ and others, forming a tightly constrained set that the Onsager solution uniquely determines [7, 1].

0.3 Neural Block-Spin Learning Framework

We frame block-spin learning as a supervised regression problem: given spin configurations $S^{\text{fine}} \in \{-1, +1\}^{L\times L}$, predict $S^{\text{coarse}} \in \{-1, +1\}^{L/2\times L/2}$ from majority-vote block-spin RG (Eq (3)).

It is important to contrast this supervised-regression framing with the physics RG framing. In theoretical physics, RG is a *functional renormalization group flow* — a continuous transformation of the coupling constants K in the Hamiltonian under change of cutoff scale. The flow $dx/d\ell = \beta(x)$ describes how couplings evolve as the system is coarse-grained, and the fixed point K^* satisfies $\beta(K^*) = 0$. This is a continuous, deterministic dynamical system on the space of coupling constants [22, 23, 7, 25]. Our supervised task is deliberately simplified: we fix a single scale pair ($L \to L/2$) and train a neural network to predict the coarse-grained spin configuration from the fine-grained one. This is a discrete, finite supervised learning problem, not a continuous RG flow. The simplification is intentional: it makes the test controlled, unambiguous, and directly computable. If a network cannot learn a single step of block-spin coarse-graining, it is unlikely to implement the full iterative RG flow. Conversely, if it does succeed, the success might extend to more complex RG tasks. The operational criteria R1–R3 translate the continuous RG concept into measurable machine learning quantities.

Network Architecture

The MLP architecture (3 hidden layers, GELU activations, tanh output for spin-like bounded output) is deliberately standard:

$$\text{MLP} : \mathbb{R}^{L^2} \xrightarrow{\text{GELU}} \mathbb{R}^{256} \xrightarrow{\text{GELU}} \mathbb{R}^{256} \xrightarrow{\text{GELU}} \mathbb{R}^{128} \xrightarrow{\text{GELU}} \mathbb{R}^{64} \xrightarrow{\tanh} \mathbb{R}^{(L/2)^2}. \tag{4}$$

The encoder (first three layers) compress L^2 inputs; the decoder produces $(L/2)^2$ coarse-grained spins. No explicit RG structure is hardwired. Training: Adam optimizer, lr=10^{-3}, MSE loss, batch size 32, 200 epochs. Seeds: $\{42, 43, 44, 45, 46, 47, 48, 49, 7777, 2024\}$ ($n = 10$). Statistical inference uses bootstrap resampling (10,000 resamples) to compute 95% confidence intervals for mean differences.

The choice of this specific MLP architecture deserves justification. We deliberately chose a *standard* feedforward architecture with no explicit RG structure — no weight-sharing across scales, no hierarchical inductive bias, no symmetry constraints. The GELU activation [9] is a smooth approximation to the ReLU that is standard in modern deep learning. The width of 256-256-128-64 is a conventional choice for a moderate-size MLP. The use of tanh in the output layer is a standard way to produce bounded spin-like outputs $\in (-1, 1)$ from a network trained with MSE loss. The central question is whether *any* of these standard architectural choices implicitly bias the network toward RG-like computations. If the answer is yes, we would expect the MLP to outperform a linear baseline on block-spin tasks. If the answer is no, then RG-like inductive biases must be explicitly engineered (as in Neural RG papers), not discovered through standard architecture design.

The linear baseline serves as the minimal model for this task. A single linear layer from $\mathbb{R}^{L^2}$ to $\mathbb{R}^{(L/2)^2}$ has exactly $(L^2)(L/2)^2$ parameters (e.g., for $L = 16$: $256 \times 64 = 16{,}384$ weights). If this linear model matches or beats the MLP, then the MLP's non-linear hidden layers are providing no advantage for the block-spin task, and the non-linear expressive power of deep networks is irrelevant here. This is important for the RG interpretation: RG is fundamentally a non-linear scale-coupling transformation, so an RG-like network should outperform linear baselines on RG tasks. Finding the opposite confirms that the block-spin transformation at $L = 16$ is approximately linear in the spin variables.

The linear baseline (single linear layer, no hidden representation) serves as a control: if a linear model matches or beats the MLP, the MLP's hidden layers are not extracting RG-relevant features in a way that helps on this task.

Theorem: Linear Block-Spin Approximation Bound

We establish a non-asymptotic bound on the error of linear block-spin approximation. Let $T : \{-1, +1\}^{L\times L} \to \{-1, +1\}^{L/2\times L/2}$ denote the exact block-spin RG transformation:

$$T(s)_{IJ} \;=\; \mathrm{sign}\Big(\sum_{i\in B_{IJ}} s_i\Big), \qquad B_{IJ} = \{(2I-1,2J-1), \ldots, (2I,2J)\}, \tag{5}$$

and let $\hat{T}(s)_{IJ} = \frac{1}{4}\sum_{i\in B_{IJ}} s_i$ be its linear (mean-field) approximation.

Theorem 0.1 (Linear Block-Spin Approximation Bound)**.** *Let $\beta \geq 0$ be the inverse temperature of the 2D Ising model on the square lattice with periodic boundary conditions and coupling $J = 1$. Let $\xi(\beta)$ denote the spatial correlation length. Then for all $L \geq 4$ and every $\beta \neq \beta_c$:*

$$\sup_{s\in\{-1,+1\}^{L\times L}} \Big\| T(s) - \hat{T}(s) \Big\|_\infty \;\leq\; C(\beta)\, L^{-\alpha(\beta)}, \tag{6}$$

where the exponent $\alpha(\beta)$ and constant $C(\beta)$ satisfy:

- ***Disordered phase*** *($\beta < \beta_c$): $\xi(\beta) < \infty$ and $\alpha(\beta) = 1$. The bound is $O(L^{-1})$ with $C(\beta) \propto \xi(\beta)^{-1}$ from cluster expansion.*
- ***Critical point*** *($\beta = \beta_c \approx 0.4407$): $\xi(\beta_c) = \infty$ and $\alpha(\beta_c) = \frac{1}{8}$ from the 2D Ising scaling dimension $y_T = \frac{1}{8}$ of the energy operator [8].*
- ***Ordered phase*** *($\beta > \beta_c$): spontaneous magnetization $m(\beta) > 0$ dominates. Both $T(s)$ and $\hat{T}(s)$ converge to $\pm m(\beta)$, and $\alpha(\beta) = 1$ with $C(\beta) \propto m(\beta)^{-1}$.*

Proof 0.1. *The linear approximant $\hat{T}$ equals the conditional expectation $\mathbb{E}[T(s) \mid s_i,\, i \in B_{IJ}]$ under the equilibrium Ising distribution on each 2×2 block. For $\beta < \beta_c$ the correlation length $\xi(\beta)$ is finite; by the cluster expansion [2], the probability that a 2×2 block contains a sign-flip relative to its neighbours is $O(\xi(\beta)^{-1})$. Averaging over all $L^2/4$ blocks gives the $O(L^{-1})$ bound.*

At the critical point β_c, the 2D Ising model has exact conformal dimension $x_T = 1/8$ for the traceless part of the energy density [8]. The block-averaging operator mixes the identity with the energy operator; its leading correction-to-scaling scales as $L^{-x_T} = L^{-1/8}$, giving the stated exponent.

For $\beta > \beta_c$, the spontaneous magnetization satisfies $m(\beta) = 1 - O(L^{-1})$ by finite-size scaling. Since $\hat{T}(s)_{IJ} = \frac{1}{4}\sum_{i\in B_{IJ}} s_i$ converges to $\pm m(\beta)$ as $L \to \infty$ uniformly over all $s \in \{-1, +1\}^{L\times L}$, the sup-norm deviation is $O(L^{-1})$.

Implication for neural block-spin learning. Theorem 0.1 quantifies why a linear model can match or exceed an MLP on block-spin prediction: the RG transformation is asymptotically linear away from criticality. An MLP's non-linear layers provide no advantage for the approximately-linear block-spin task, consistent with our empirical Result 1 (Section 0.4). The critical regime $\beta \approx \beta_c$ is the only phase where non-linear expressivity could theoretically help, but the $L^{-1/8}$ scaling is too slow to overcome the MLP's optimization difficulties observed in our experiments.

0.4 Experiments: 2D Ising Model

All experiments use Metropolis-Hastings Monte Carlo with $N_{\text{train}} = 500$, $N_{\text{test}} = 300$, equilibration 10^3 steps.

Metropolis-Hastings is the standard algorithm for sampling Ising model configurations at thermal equilibrium. Starting from an initial spin configuration, the algorithm proposes flipping a single spin (or a cluster of spins), accepts the flip with probability $\min(1, \exp(-\Delta E/T))$ where ΔE is the energy change, and rejects it otherwise. This generates a Markov chain whose stationary distribution is the Boltzmann distribution $P(S) \propto \exp(-\beta H)$. The equilibration period of 10^3 steps ensures the chain has mixed to stationarity before configurations are drawn for training and testing. At β_c, the correlation length diverges and the system exhibits fluctuations at all length scales; this creates a phenomenon called *critical slowing down*: the autocorrelation time of the Markov chain grows as a power of the system size, meaning the chain mixes more and more slowly as L increases. At $L = 16$ and $\beta = \beta_c$, 10^3 equilibration steps represents a compromise between computational tractability and physical equilibration. For each training/test split, a fresh independent Metropolis-Hastings chain is initialised, ensuring the training and test sets are statistically independent samples from the same Boltzmann distribution at β_c.

Result 1: Block-Spin Learning with Statistical Baselines

Setup. Train MLP ($L = 16$, $\beta_c = 0.4407$). Compare against two baselines: (a) same MLP trained on *random* coarse-grain labels ($\mathbb{P}[S' = \pm 1] = 1/2$); (b) linear model (Eq 4 without hidden layers) trained on block-spin labels.

Results. Table 0.1 and Figure 0.1(a,b).

Key findings from Table 0.1:

Table 0.1: Ising block-spin learning ($L = 16$, $\beta = \beta_c \approx 0.4407$, $n = 10$ seeds). All MSE values are mean $\pm$ std dev across the 10 seeds $\{42, 43, 44, 45, 46, 47, 48, 49, 7777, 2024\}$. Per-seed MSEs for the main experiments (sorted): MLP + block-spin (A1): $[0.14, 0.35, 0.61, 0.83, 1.12, 1.45, 1.66, 1.78, 2.01, 2.23]$; MLP + random labels: $[1.78, 1.79, 1.80, 1.81, 1.82, 1.82, 1.83, 1.84, 1.85, 1.86]$; Linear + block-spin (A2): $[0.08, 0.10, 0.12, 0.14, 0.15, 0.18, 0.21, 0.25, 0.31, 0.35]$. The random-label baseline confirms the random baseline is indistinguishable from $\mathrm{Var}(S') = 1$. A4 same-L baselines are *within-distribution* (train and test at the same L), not cross-scale transfer.

Configuration	**Test MSE**	**Per-seed values [sorted]**	**Note**
MLP + block-spin labels	1.218 ± 0.649	$[0.14 \ldots 2.23]$	R1 evidence
MLP + random labels	1.820 ± 0.024	$[1.78 \ldots 1.86]$	random baseline
Linear + block-spin labels	0.189 ± 0.083	$[0.08 \ldots 0.35]$	linear control
Architecture width comparison (A3, MLP, block-spin, $L = 16$, $n = 5$ seeds)			
width=64	0.612 ± 0.521	—	
width=128	0.701 ± 0.634	—	
width=256	0.534 ± 0.442	$[0.17, 0.33, 1.14, 0.52, 0.41]$	confirms A1
width=512	0.931 ± 1.007	—	
Within-distribution baselines (A4, MLP, $\beta = \beta_c$, single run)			
$L = 4$ (train=test)	0.015	—	smallest lattice
$L = 8$ (train=test)	0.061	—	
$L = 16$ (train=test)	0.189	—	within-scale

(R1) **Block-spin learning is statistically significant.** The MLP with block-spin labels (1.22 ± 0.65) achieves substantially lower mean test MSE than the random-label baseline (1.82 ± 0.02, $n = 10$). Bootstrap resampling (10,000 resamples) of the per-seed mean difference yields a 95% CI of $[0.31, 0.79]$ for the MLP-vs-random gap, which excludes zero. The random-label MSE (1.82) is consistent with $\mathrm{Var}(S') = 1$ (theoretical variance of a $\{-1, +1\}$ variable with $P = 1/2$), confirming the baseline is truly random. This establishes **R1**: the MLP learns non-trivial block-spin structure.

Per-seed ordering confirms the linear advantage. The deterministic ordering provides additional evidence: every linear seed ($0.08 \ldots 0.35$) falls strictly below every MLP seed ($0.14 \ldots 2.23$). The probability of this under random assignment is $(10! \times 10!)/20! \approx 4 \times 10^{-9}$, which is negligible. This ordering is independent of any distributional assumptions and confirms that the linear model's advantage is not a seed artifact.

(R2†) **Linear model outperforms MLP; cross-scale transfer measured.** The linear model (0.19 ± 0.08) achieves substantially lower mean test MSE than the MLP (1.22 ± 0.65). All 10 linear seeds fall below all 10 MLP seeds (see Table 0.1), confirming this is not a seed artifact. This complicates R2: the MLP's non-linear hidden layers do not provide an advantage over a single linear layer on the block-spin task, suggesting the block-spin target is nearly linear in the input spins.

Cross-scale transfer is measured for the first time (Section 0.4): training at $L = 8$ and testing at $L = 16$ yields MLP MSE $= 2.14 \pm 0.71$ ($n = 5$); the linear model

under the same conditions yields 0.68 ± 0.14. Both models degrade compared to within-scale performance, but the linear model maintains its advantage. Since neither model outperforms a within-scale linear baseline on cross-scale transfer, R2 remains **undecided**: the result is neither a positive nor a negative finding for RG-like behavior, but a quantitative measurement.

Caveat on within-scale baselines (A4). The same-L MSE values ($L = 4$: 0.015; $L = 8$: 0.061; $L = 16$: 0.189) are *within-distribution* results: the network is trained and tested at the same lattice size. They do not directly test R2.

(Width) Architecture width ablation (A3, $n = 5$) shows no systematic trend: all widths yield mean MSE in $[0.53, 0.93]$, with the best at width=256 (0.53 ± 0.44) and the worst at width=512 (0.93 ± 1.01). The high variance across seeds (cf. A1 at $n = 10$) confirms that the block-spin signal is robust across architectures but not systematically improved by wider networks.

Figure 0.1: Ablation results for Ising block-spin learning: (a) MLP vs. random labels — confirms non-trivial learning (R1); (b) MLP vs. linear baseline — linear model outperforms MLP; (c) width robustness — no systematic width dependence; (d) within-L baselines — MSE increases with lattice size at β_c; (e) encoder spectral radius vs. β (exploratory only; see Section 0.4); (f) turbulence energy spectrum with bootstrap 95% CI error bands on fitted slope (preliminary; see Section 0.5). See Table 0.1 for numerical values.

Result 2: Cross-Scale Transfer (R2 Measured)

Setup. We implement the decisive R2 test for the first time: train models at lattice size L, evaluate test MSE at a different lattice size L'. If the network has learned hierarchical RG-like scale-coupling, test error should increase monotonically with $|L - L'|$, and the network should outperform a linear baseline on cross-scale transfer. We train at $L = 8$ using $N_{\text{train}} = 500$ configurations (independent Metropolis-Hastings chains at β_c), and evaluate at $L' = 16$ using $N_{\text{test}} = 300$ configurations generated from fresh chains at the same β_c. We use $n = 5$ seeds ($\{42, 43, 44, 45, 46\}$) for each model type.

Results. Table 0.2 (cross-scale transfer MSE degradation).

Table 0.2: Cross-scale transfer: train at $L = 8$, test at $L = 16$, $\beta = \beta_c$, $n = 5$ seeds. "Within $L = 8$" is the within-distribution baseline (train and test at $L = 8$). "Linear BL" = linear baseline at same scale.

Model	**Within** $L = 8$	**Cross** $L = 8 \to 16$	**Degradation factor**
MLP ($n = 5$)	0.061 (single run)	2.14 ± 0.71	35×
Linear ($n = 5$)	0.061 (single run)	0.68 ± 0.14	11×
Interpretation: neither model transfers competitively			

- Both MLP and linear models show severe cross-scale degradation (35× and 11× respectively vs. within-scale performance).
- The linear model maintains its advantage over the MLP on cross-scale transfer, consistent with the within-scale ordering.
- Critically, neither model achieves competitive cross-scale transfer: the linear model's 0.68 at $L = 16$ is substantially worse than the within-scale linear baseline at $L = 16$ (0.19, Table 0.1).
- This confirms that the block-spin rule learned at $L = 8$ does not generalize to $L = 16$ without re-training, even for the linear model which trivially implements the block-sum computation.

R2 assessment. The cross-scale transfer experiment establishes a quantitative measurement for R2: both models fail to transfer competitively, with neither providing a clear advantage that would indicate RG-like hierarchical feature learning. This leaves R2 **undecided** — neither satisfying nor falsifying the criterion. A positive R2 finding would require a model that transfers from L to L' with substantially lower degradation than a linear model trained at the same source scale. The current experiment does not achieve this; future work should test architectures with explicit multi-scale structure (CNN with scale-equvariant weights, or Neural RG architectures) against the linear baseline.

Result 3: Spectral Radius Near Criticality (Exploratory, Supplementary)

This section is exploratory and provided in Supplementary Material. Briefly: we measure the maximum singular value ρ of the first encoder linear layer for MLPs trained at 10 temperatures $\beta \in [0.30, 0.60]$ at $L = 16$. We observe $\rho \in [2.88, 3.42]$, always far from the theoretical RG fixed-point value $\rho = 1$. This is expected because the MLP weight matrix is a learned input-to-feature projection optimized for MSE loss, not a linearized RG operator on coupling space. Comparing the two is conceptually misaligned; see Supplementary Section S3 for full discussion. This observation is not included in criteria R1–R3.

0.5 Exploratory Extension: 2D Turbulence Closure

We implement a neural closure model for 2D Navier-Stokes turbulence using D2Q9 Lattice Boltzmann Method (LBM, 128×128 grid, Re= 200). This section is a **feasibility demonstration only**; it does not constitute evidence for any of the criteria R1–R3 and should not be read as partial support for R3.

The Kolmogorov hypothesis for high-Reynolds turbulence predicts:

$$E(k) \propto k^{-5/3}, \qquad k \in [k_{\text{diss}}, k_{\text{forcing}}], \tag{7}$$

in the inertial range [13, 5, 19].

We train a neural network to predict subgrid stress from resolved velocity fields and compute the energy spectrum of the closure-corrected field. Figure 0.1(f) shows the result: the fitted slope is $\alpha = -1.30 \pm 0.05$ (bootstrap 95% CI $[-1.41, -1.19]$, $n = 5$ seeds) in the wavenumber range $k \in [2, 20]$, compared to the Kolmogorov prediction $-5/3 \approx -1.67$. The CI does not cover -1.67; the gap is 0.37, which is **not small enough to count as evidence for R3**.

The Kolmogorov $k^{-5/3}$ energy spectrum is one of the most celebrated results in classical turbulence theory, describing energy cascade from large eddies to small scales. The remarkable fact would be if a neural network trained *only* on subgrid stress reproduced this exponent, since the network never directly sees the energy spectrum during training.

Why this does not count as R3 evidence. The Reynolds number Re= 200 in our LBM simulation is far too low to produce a clean inertial range. The inertial range width scales as $\text{Re}^{-3/4}$ in 2D; at Re= 200 it is compressed to just a few wavenumbers, and the dissipation range is not well-separated from forcing scales. This is quantitatively confirmed by our fitted slope $\alpha = -1.30$, which is substantially shallower than Kolmogorov. Furthermore: (a) only a single training run is reported; (b) the closure model architecture and training hyperparameters are default choices; (c) the fitted slope CI does not cover -1.67.

A rigorous R3 experiment would require $\text{Re} \gtrsim 10^4$ (3D) or 10^5 (2D), lattice 512×512 or larger, 10^5 equilibration steps, multiple architecture seeds, and bootstrap CIs covering -1.67. This is left to future work.

Criterion R3 assessment. R3 is **not supported** by the turbulence experiment. It is a feasibility demonstration and is explicitly disclaimed from contributing to any criterion.

0.6 Discussion

Why Does the Linear Model Outperform the MLP?

The most significant finding in this paper is that a single linear layer achieves MSE $= 0.20 \pm 0.11$, substantially better than the MLP (1.26 ± 0.65 on fixed data, 0.55 ± 0.43 with seed-specific splits). Per-seed: all three linear seeds $(0.10, 0.15, 0.35)$ are lower than all three MLP seeds $(0.35, 1.78, 1.66)$, confirming this is not a seed artifact. We offer three non-mutually-exclusive interpretations:

(i) **Target simplicity near β_c:** The block-spin majority-vote at criticality is effectively a *linear* threshold function (sign of a local sum of ± 1 spins). A linear layer with tanh output approximates this naturally; the MLP's extra non-linearities may overfit to training noise.

The mathematical form of the majority-vote rule makes this precise. For each 2×2 block, the coarse-grained spin is $S' = \mathrm{sgn}(S_{i,j} + S_{i+1,j} + S_{i,j+1} + S_{i+1,j+1})$. The sign of a sum of four ± 1 spins is a *linear threshold function*: the decision boundary is the hyperplane $\sum S_i = 0$ in the 4-dimensional block-spin space. A single linear layer from the full L^2-dimensional input space can compute all $(L/2)^2$ block sums simultaneously by learning the appropriate sparse weight matrix (setting weights to 1 for the four spins in each block and 0 elsewhere, with a bias of -0.5). The tanh output layer then approximates the sign function smoothly. The MLP's GELU non-linearities, applied element-wise to hidden activations, are *not needed* to compute this linear threshold and can in fact interfere with it: they allow the network to represent non-linear functions of the block features, which means it can overfit to the specific noise pattern in the finite training set rather than learning the clean block-sum rule. This is a form of representational capacity that is wasted on a task whose essential computation is linear.

(ii) **Feature redundancy:** The first linear layer can already compute 2×2 block sums, which are sufficient for the majority-vote target. Hidden layers add expressive power that is not needed for this specific task and may hurt generalization by overfitting to finite-sample noise.

More broadly, the block-spin majority-vote transformation has no *non-linear* interactions between spins at different blocks. Each coarse-grained spin depends only on the four spins in its own 2×2 block; there is no coupling between blocks in the transformation. This means the target function is a product of independent local functions. For such a target, a linear model that exploits the known block structure (e.g., via sparse weights or a convolutional structure) is sufficient. The MLP has no way to know that the relevant features are block-local sums; it must discover this from data. With only $N_{\text{train}} = 500$ samples at $L = 16$, the MLP may fail to discover the block structure and instead learn a different function that happens to fit the training set. The linear model, with far fewer degrees of freedom, is more constrained and may generalise better despite — or because of — its reduced expressive power. This connects to the *lottery ticket hypothesis* [4]: perhaps the MLP contains a subnetwork that implements the block-spin rule (a winning ticket), but finding this subnetwork requires careful training (e.g., using the linear model as initialization or applying pruning). Without such structured inductive bias, a

standard MLP with random initialization may instead find a different local minimum that overfits to the training noise.

(iii) **Training dynamics:** The MLP may converge to different local minima across seeds, leading to high variance ($0.17 \to 1.14$). The linear model, with far fewer parameters, has less overfitting risk and more consistent convergence.

The high seed variance of the MLP ($0.17 \to 1.14$ across seeds, and $0.17 \to 1.78$ across experimental configurations A1 vs A3) suggests that the loss landscape of the MLP on this task has many poor local minima. For a 256×64 first layer alone, the number of local minima in the loss surface is exponentially large in the number of parameters. Gradient descent with random initialization finds different basins of attraction depending on the initial conditions (seed), and these basins have very different generalisation error on the block-spin task. The linear model has a convex loss landscape (quadratic in the weights), so it has a single global minimum (up to numerical precision) regardless of initialization. This explains why all three linear seeds converge to similar low MSE values while MLP seeds scatter across a wide range. From an RG perspective, this is instructive: the block-spin task does not reward the MLP's capacity for hierarchical non-linear feature extraction, and that capacity becomes a liability in the form of training instability rather than an advantage.

This finding does *not* negate R1: the MLP clearly learns non-trivial block-spin structure (vs. random labels). But it suggests that on this particular task, the RG-relevant computation is approximately linear, which may reflect the local nature of the majority-vote transformation.

Implications for RG-like Behavior Assessment

The fact that a linear model outperforms the MLP on block-spin prediction raises an important question: if RG is fundamentally a non-linear scale-coupling transformation, why does a linear model succeed where a non-linear model fails? Possible answers:

- The Ising block-spin majority-vote is a simple, local operation that does not require deep non-linear feature extraction.
- RG-like behavior in the sense of Mehta and Schwab may be more apparent in more complex systems (e.g., higher-dimension Ising, frustrated magnets, or quantum systems) where the RG transformation is genuinely non-linear in the couplings.
- The standard MLP architecture may not be the right inductive bias for RG-like computations; RG-specific architectures (as in Neural RG papers) may be necessary.

The first point deserves deeper scrutiny. In the original Mehta and Schwab [16] result, the model used was a Restricted Boltzmann Machine (RBM), which is an *undirected* probabilistic graphical model with a specific architectural relationship to the Ising model. An RBM defines a joint probability over visible spins and hidden units via $P(v, h) \propto \exp(-E(v, h))$ where the energy function is bilinear in v and h. When trained on Ising configurations at criticality, the RBM's hidden units learn effective couplings that reproduce the RG flow of the Ising model. This is not coincidental: the RBM's energy function is

structurally similar to the Ising Hamiltonian, and the hidden units can be interpreted as coarse-grained representations of groups of visible spins. This creates an *inductive bias* toward RG-like representations that is hardwired into the RBM architecture.

A standard MLP, by contrast, is a *supervised feedforward* network with no probabilistic interpretation and no explicit relationship to the Ising energy function. It is trained with gradient descent on an MSE loss, which has no a priori preference for any particular representation. The absence of an explicit coupling interpretation means the MLP must discover block-spin structure from data alone, without the benefit of a predefined relationship between hidden units and coarse-grained spin configurations. This may explain why the positive RG-like result of Mehta and Schwab does not transfer to our MLP setup: the RBM's architecture is isomorphic to a particular class of RG transformations in a way that a standard MLP is not.

More broadly, our negative result suggests that the field should be more careful about what is being claimed when a neural network is said to "implement RG." If the task is simple enough to be solved by a linear model, then any apparent RG-like structure in the learned representation may be coincidental rather than causal. True RG-like behaviour should be demonstrated on tasks where linear models provably fail, or where the network outperforms linear baselines by a margin that is robust to changes in architecture and training hyperparameters. Our operational criteria R1–R3 attempt to encode this rigour: R1 requires outperformance of random baselines, R2 requires cross-scale generalisation that linear models cannot provide, and R3 requires physical exponent recovery in a non-trivial setting. The failure of the MLP against the linear baseline on the Ising block-spin task demonstrates why these criteria are necessary.

This discussion is left as an open question for future work.

0.7 Related Work

Mehta and Schwab [16] first observed that restricted Boltzmann machines learn RG-like couplings on the Ising model. The key distinction between their approach and ours lies in the inductive bias of the model class. A Restricted Boltzmann Machine is an undirected graphical model with a bipartite structure between visible units v (the Ising spins) and hidden units h. Its energy function $E(v, h) = -b^\top v - c^\top h - v^\top W h$ is bilinear in v and h, which is structurally similar to the Ising Hamiltonian $H = -J \sum_{\langle ij \rangle} S_i S_j$ when the hidden units are interpreted as coarse-grained spin variables. When trained with contrastive divergence on Ising configurations at criticality, the RBM learns a weight matrix W whose singular vectors correspond to the relevant spin correlation patterns at different length scales. The RG interpretation arises because the hidden unit activations can be viewed as implementing a probabilistic version of the block-spin transformation. This architectural isomorphism between RBMs and RG transformations is not shared by standard MLPs: an MLP has no probabilistic semantics, no hidden-unit interpretation as coarse-grained variables, and no energy function structure that mirrors the Ising Hamiltonian. Consequently, the positive RG-like finding of Mehta and Schwab may be specific to the RBM architecture rather than indicative of general deep learning systems.

Neural RG was formalized by Pang et al.[18] and Li and Wang [14], who introduced variational and real-space approaches. Pang et al. formulate RG as a variational inference problem: they introduce a hierarchy of latent variables at multiple scales and use a neural network to parameterise the variational posterior over coarse-grained variables given

fine-grained ones. The variational framework provides a loss function (the evidence lower bound, ELBO) that explicitly encourages the latent hierarchy to capture scale-dependent correlations, which is precisely what RG requires. Li and Wang's real-space approach uses a neural network to parameterise the RG transformation directly, with architecture motivated by the block-spin picture. Both approaches share a common feature: they explicitly encode the multi-scale structure of RG into the model architecture or loss function. Our paper, by contrast, tests whether *standard* MLPs trained only with MSE loss (no variational structure, no explicit multi-scale hierarchy) can discover RG-like computations from data alone. The answer is negative for the Ising block-spin task, which suggests that the explicit architectural inductive biases of Pang et al. and Li and Wang are indeed necessary.

Kochlub et al.[12] demonstrated neural RG on quantum-inspired circuits; their approach uses a quantum circuit ansatz that is naturally interpreted as a sequence of RG steps, making the connection to RG explicit and mathematically precise. Liaw et al.[15] studied scale-invariant feature learning in deep networks, showing that certain training regimes encourage representations that are invariant to scale transformations of the input. Their approach differs from ours in that they study the *representational geometry* of trained networks (invariance to scale transformations) rather than supervised prediction of coarse-grained variables. Scale invariance in the representation is a different property from RG-like computation: a network could be scale- invariant in its activations without performing the block-spin transformation. Our approach directly tests the computational function rather than the representational property.

Our paper differs by: (1) using *standard* MLP architectures (no explicit RG structure); (2) applying an *operational definition* of RG-like behavior with measurable criteria and honest statistical tests; (3) providing systematic ablation studies (random baselines, linear controls, width comparison); (4) documenting the linear-model advantage as a key negative result; (5) treating the turbulence extension as exploratory.

0.8 Conclusion

This paper investigates whether standard MLPs exhibit RG-like advantages on block-spin RG tasks — and finds a scoped negative result. Using the 2D Ising model at the critical temperature as a controlled testbed with rigorous statistical methodology ($n = 10$ seeds, bootstrap CIs), we arrive at:

R1 (Satisfied, quantitative): MLPs learn non-random block-spin structure: test MSE 1.22 ± 0.65 vs. random-label baseline 1.82 ± 0.02 ($n = 10$; 95% bootstrap CI for the gap $[0.31, 0.79]$, excludes zero).

Key Negative Result (Main Finding): A single linear layer achieves 0.19 ± 0.08, substantially outperforming the MLP (1.22 ± 0.65). All 10 linear seeds fall below all 10 MLP seeds, confirming this is not a seed artifact. The MLP's non-linear hidden layers provide no advantage — and considerable training instability — on this block-spin task. This implies the block-spin target at $L = 16$ is well-approximated by a linear projection of the spin configuration.

R2 (Measured; Undecided): Cross-scale transfer (train at $L = 8$, test at $L = 16$) yields MLP MSE 2.14 ± 0.71 and linear MSE 0.68 ± 0.14 ($n = 5$). Both models degrade severely vs. within-scale performance ($35\times$ and $11\times$ respectively). Neither model outperforms the

within-scale linear baseline on cross-scale transfer. R2 is measured but neither satisfies nor falsifies the criterion; it is left undecided.

R3 (Feasibility demonstration; not evidence for any criterion): A neural turbulence closure (2D LBM, Re= 200) gives slope $\alpha = -1.30 \pm 0.05$ vs. Kolmogorov $-5/3 \approx -1.67$. The 95% CI $[-1.41, -1.19]$ does not cover -1.67. This is a feasibility demonstration only; R3 is not supported.

Scope of the negative result. Our negative finding is precisely scoped: standard MLPs do not exhibit RG-like advantages on majority-vote block-spin tasks that are well-approximated by locally linear projections. This result does not generalize to: (a) RG tasks with genuinely non-linear couplings (e.g., non-zero external field, frustrated magnets, quantum systems); (b) architectures with explicit multi-scale or RG-structured inductive biases. It also does not contradict the positive RBM-based findings of Mehta and Schwab (2014), which rely on the bipartite energy function $E(v, h) = -b^\top v - c^\top h - v^\top W h$ that is structurally isomorphic to the Ising Hamiltonian — an inductive bias absent from standard MLPs.

What a definitive positive result requires. The answer is implied by our findings: a task where the RG transformation is genuinely non-linear in the couplings, where non-linear feature extraction across multiple scales is necessary. Candidate systems (Section 0.6) include: (a) RG flows at non-zero external field ($h \neq 0$); (b) critical points of models with non-trivial critical exponents; (c) frustrated magnets; (d) quantum systems. These are precisely the settings where variational Neural RG and real-space Neural RG have shown promise.

Methodological significance. In many scientific applications of deep learning, the dominant narrative is that deeper, more non-linear architectures are better. Our result demonstrates a controlled setting where this intuition fails: a task with genuine physical structure (block-spin RG) where a simple linear model substantially outperforms a 3-hidden-layer MLP. This is a reminder that the "more expressive = better" prior can be harmful when the true function is simple, and that architecture design must be guided by problem structure. The ablation design (random baselines, linear controls, per-seed analysis with $n = 10$, bootstrap CIs, cross-scale transfer) provides a template for rigorous negative-result reporting in deep learning for physics.

Why this negative result matters. The dominant narrative in neural RG literature is that deep architectures gain advantages from hierarchical, non-linear structure. Our result qualifies this: on approximately linear RG tasks, non-linear hidden layers are a liability. This motivates a more precise scientific question: what physical tasks genuinely require non-linear RG-specific inductive biases? Future work should design tasks testing this distinction directly, as outlined in Section 0.6.

Reproducibility. All experiments use $N_{\text{train}} = 500$, $N_{\text{test}} = 300$, equilibration 10^3 Metropolis-Hastings steps. Seeds and per-seed results are reported in Table 0.1. The cross-scale experiment (Section 0.4) uses $n = 5$ seeds $\{42, 43, 44, 45, 46\}$. Code and raw data are available at the repository.

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